BASICS OF FIM · WI001287

Statistical Learning

1: Fundamentals statistic & probability theory

Univ.-Prof. Dr.-Ing. habil. Niklas Kühl
Lars Böcking

Chair for Information Systems and
Human-centric Artificial Intelligence
University of Bayreuth
TUM School of Management

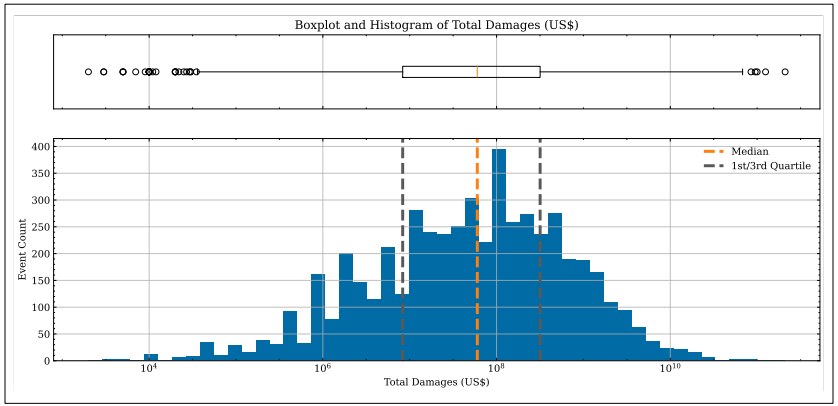


Figure 1.5: Boxplot and histogram of total damages (US$)

Statistical learning for predicting real world natural disasters.

‘Disastrous beyond comprehension’: 10,000 missing after Libya floods

Neighbourhoods washed away in port city of Derna, where two dams burst, with many bodies swept out to sea [1]

UN says most Libya flood deaths could have been avoided, as reports emerge of how 7-meter wave ‘wiped out’ buildings

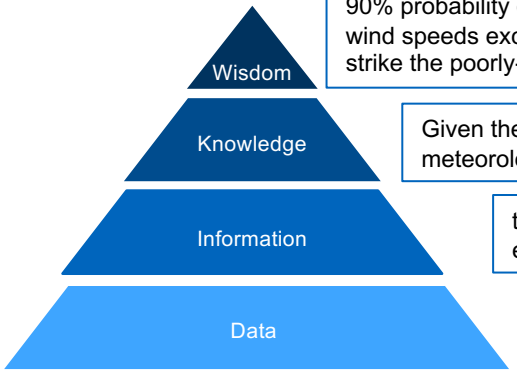
By Sana Noor Haq, Celine Alkhaldi and Eyad Kourdi, CNN
Updated 4:33 AM EDT, Fri September 15, 2023 [2]

Pre-event satellite image of Derna, Libya from July 1, 2023 - 32°46'N, 22°38'E



Via Nahel Belgherze - @WxNB_ on X - Satellite Imagery courtesy of Maxar Technologies [3]

?! What if the flood was predicted in advance?



90% probability of a catastrophic weather event, featuring 300 mm rainfall, wind speeds exceeding 100 km/h, and towering waves at 10 meters, poised to strike the poorly-prepared city of Derna, Libya, in the next 48 hours.

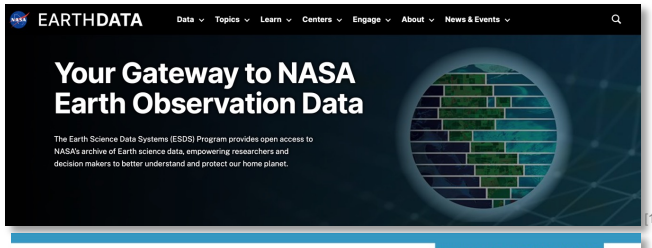
Given the urban layout maximum seeps can be modelled for meteorological scenarios

trends in monthly precipitation, summary statistics on historic events

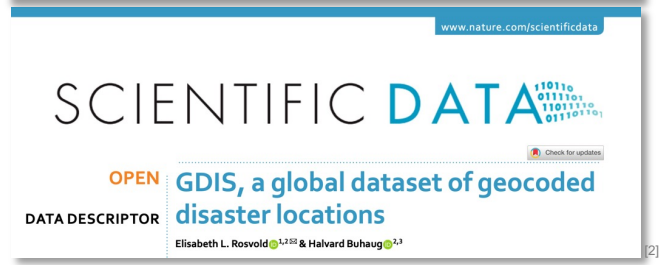
Stream: precipitation in rain gage l/m^2, Humidity sensor g/m^3, wind vane km/h

[1] https://www.theguardian.com/world/2023/sep/12/libya-floods-death-toll-dams-burst
[2] https://edition.cnn.com/2023/09/14/world/libya-floods-un-deaths-intl/index.html
[3] https://twitter.com/wxnb_?lang=de

We look at real world data from NASA throughout FIM-Basics.



[1]



[2]

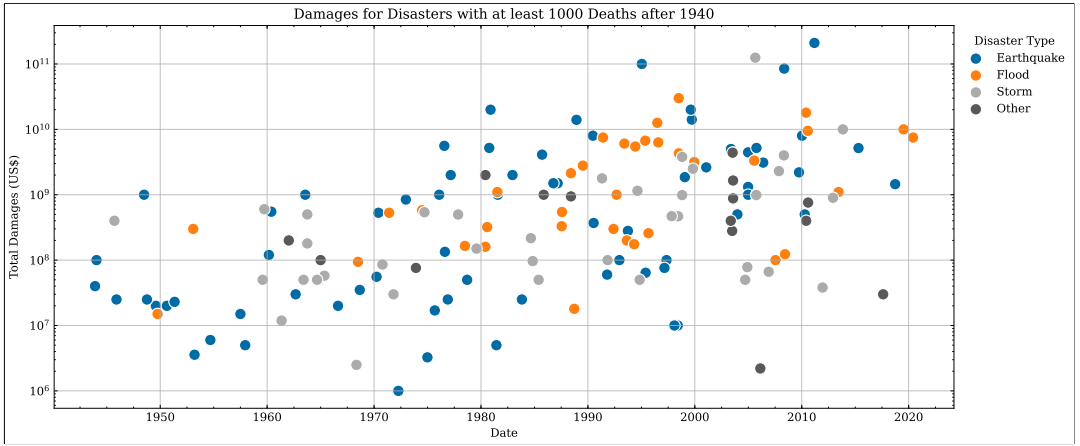


Figure 1.1: damage distribution

- Real world data from NASA on natural disasters around the world from 1900-2021 covering >16K events [3]
- Disasters such as Storm, Drought, Earthquake, Epidemic, Flood, Volcanic activity among others
- Impact reported by number of death, number of homeless, total damage, etc.

[1] https://www.earthdata.nasa.gov
[2] https://www.nature.com/articles/s41597-021-00846-6
[3] https://www.kaggle.com/datasets/brsdincer/all-natural-disasters-19002021-eosdis

We look at real world data from NASA throughout FIM-Basics.



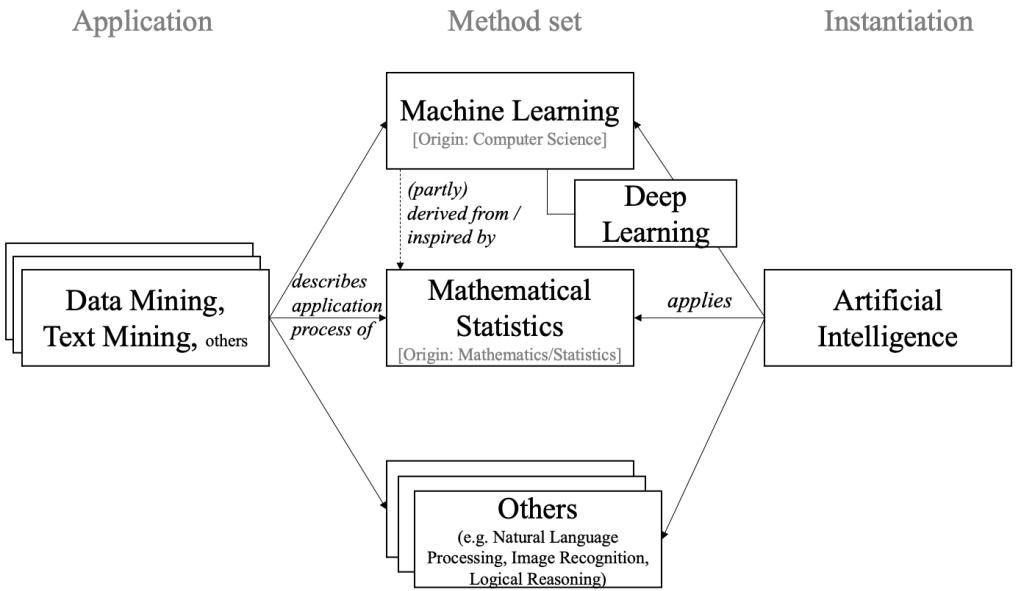
Agenda.

- Terms Artificial Intelligence & Machine Learning
- Variables types and notations
- Descriptive univariate statistics
- Bivariate statistics
- Probability theory
- Bayes Theorem

Agenda.

- Terms Artificial Intelligence & Machine Learning
- Variables types and notations
- Descriptive univariate statistics
- Bivariate statistics
- Probability theory
- Bayes Theorem

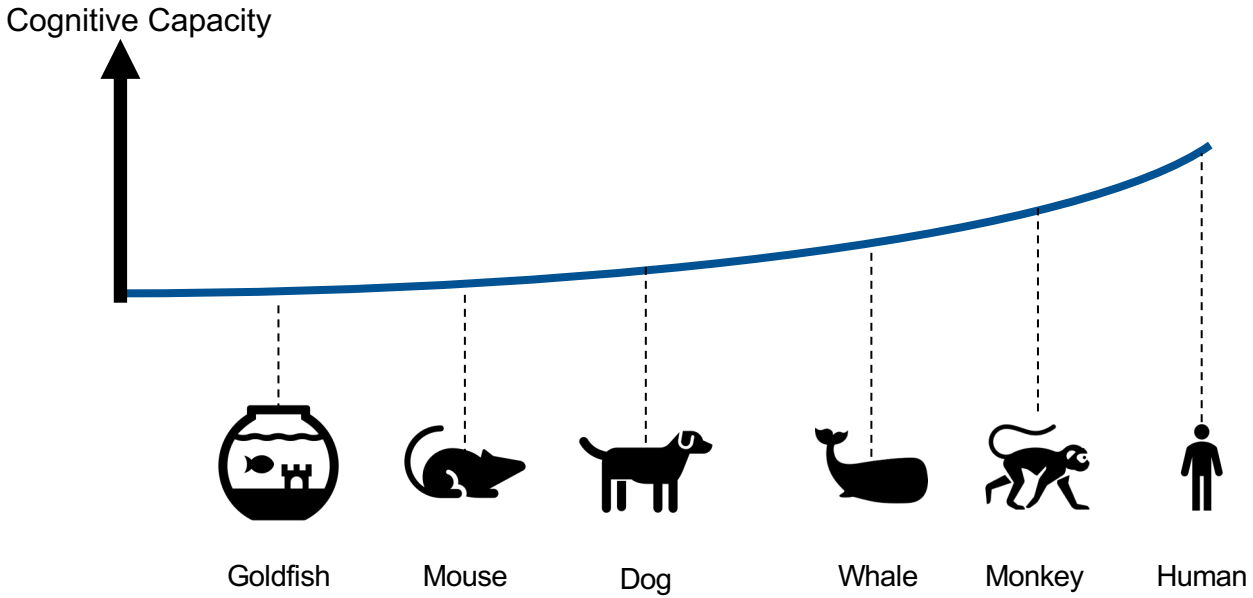
Artificial Intelligence & Machine Learning: So many terms – What do they mean?



Kühl, N., Goutier, M., Hirt, R., & Satzger, G. (2019). Machine Learning in Artificial Intelligence: Towards a Common Understanding.

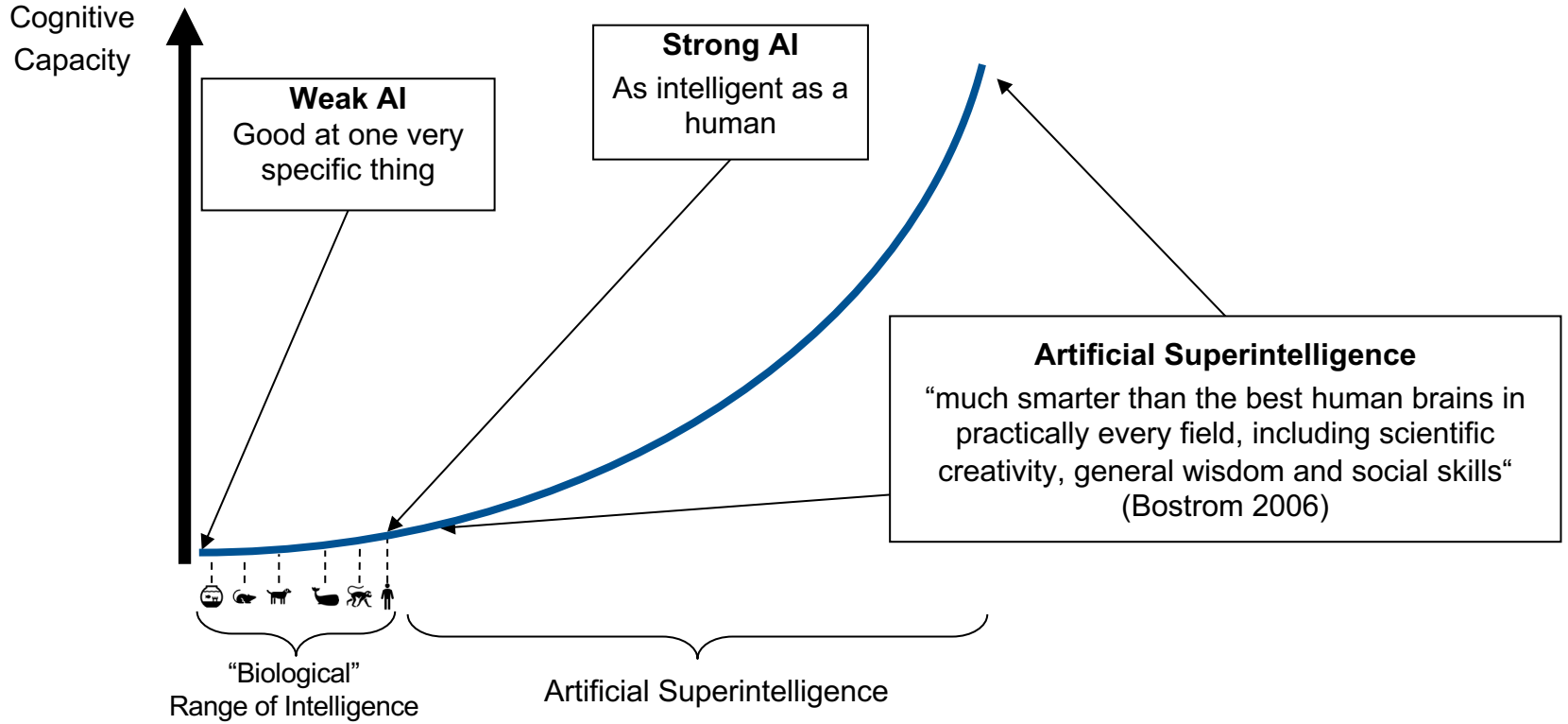
Definition of AI: An intelligence-level-oriented view.

Intelligence: *The ability to perceive information, to retain it as knowledge and to apply it towards adaptive behaviors within an environment.*



Cairo (2011)

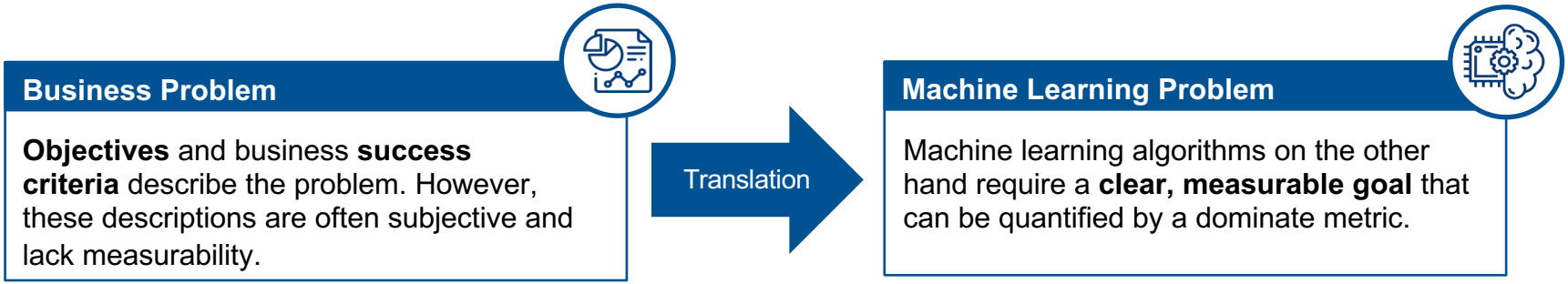
What “levels” of AI can we define?



Philosophy of this course.

- It is essential to understand the **underlying principles** behind various techniques in order to know how and when to apply them effectively.
- One must understand the **simpler methods** first in order to grasp the more sophisticated ones.
- It is important to **accurately assess the performance** of a method, to know how well or how badly it is working [simpler methods often perform as well as fancier ones!]
- This is an **exciting research area**, having important applications in science, industry, and finance.
- Statistical learning is a **fundamental ingredient** in the training of a modern **data scientist**.

Problem Statement: Machine learning algorithms need measurable goals.



Familiarize yourself with our hands-on real-world dataset.

EARTHDATA

Year	Seq	Glide	Disaster Group	Disaster Subgroup	Disaster Type	Disaster Subtype	Disaster Subsubtype	Event Name	Country	ISO	Region	Continent	Location	Origin	Associated Dis	Associated Dis2
1900	9002		Natural	Climatological	Drought	Drought			Cabo Verde	CPV	Western Africa	Africa	Countrywide		Famine	
1900	9001		Natural	Climatological	Drought	Drought			India	IND	Southern Asia	Asia	Bengal			
1902	12		Natural	Geophysical	Earthquake	Ground movement			Guatemala	GTM	Central America	Americas	Quezaltenango, San Marcos		Tsunami/Tidal wave	
1902	3		Natural	Geophysical	Volcanic activity	Ash fall		Santa Maria	Guatemala	GTM	Central America	Americas				
1902	10		Natural	Geophysical	Volcanic activity	Ash fall		Santa Maria	Guatemala	GTM	Central America	Americas				
1903	6		Natural	Geophysical	Mass movement (dry)	Rockfall			Canada	CAN	Northern America	Americas	Frank, Alberta			
1903	12		Natural	Geophysical	Volcanic activity	Ash fall		Mount Karthala	Comoros (the)	COM	Eastern Africa	Africa				
1904	3		Natural	Meteorological	Storm	Tropical cyclone			Bangladesh	BGD	Southern Asia	Asia	Chittagong			
1905	5		Natural	Geophysical	Mass movement (dry)	Rockfall			Canada	CAN	Northern America	Americas	Spence's Bridge, British Columbia			
1905	3		Natural	Geophysical	Earthquake	Ground movement			India	IND	Southern Asia	Asia	Kangra			
1906	14		Natural	Geophysical	Earthquake	Ground movement			Chile	CHL	South America	Americas	Valparaiso		Tsunami/Tidal wave	
1906	2		Natural	Geophysical	Earthquake	Ground movement			Colombia	COL	South America	Americas	Tumako		Tsunami/Tidal wave	
1906	23		Natural	Hydrological	Flood				Belgium	BEL	Western Europe	Europe	Louvain region			
1906	24		Natural	Hydrological	Flood				Belgium	BEL	Western Europe	Europe				
1906	15		Natural	Meteorological	Storm	Tropical cyclone			Hong Kong	HKG	Eastern Asia	Asia				
1907	6		Natural	Geophysical	Earthquake	Ground movement			China	CHN	Eastern Asia	Asia	Tien Shan			
1907	1		Natural	Biological	Epidemic	Bacterial disease		Bubonic	India	IND	Southern Asia	Asia				
1908	11		Natural	Geophysical	Mass movement (dry)	Landslide			Canada	CAN	Northern America	Americas	Notre Dame de La Salette, Quebec			
1909	10		Natural	Meteorological	Storm	Tropical cyclone			Bangladesh	BGD	Southern Asia	Asia	Bakerganj			
1909	13		Natural	Meteorological	Storm	Tropical cyclone			Bangladesh	BGD	Southern Asia	Asia	West Sundarbans			
1909	1		Natural	Biological	Epidemic	Bacterial disease		Bubonic	China	CHN	Eastern Asia	Asia				
1909	9		Natural	Geophysical	Earthquake	Ground movement			France	FRA	Western Europe	Europe	Lambesc, Saint-Cannat, Rognes, Vernegues, Charleval, La Roque d'Antheron (Haute-Provence)			

kaggle

? Which dynamics would you like to understand?

Agenda.

- Terms Artificial Intelligence & Machine Learning
- Variables types and notations
- Descriptive univariate statistics
- Bivariate statistics
- Probability theory
- Bayes Theorem

Differentiate features and target variables in a dataset for supervised learning.

Feature: an individual measurable property or characteristic of a phenomenon

Target: is the variable the model is trying to predict

ID	Feature 1	Feature 2	…	Feature p	Target
1					
2					
3					
4					
5					
…					
n					

Model **learns** relationships from input to a target variable on historical data….

…and then **applies** this knowledge to instances with unknown target variables.

EARTHDATA

Date	Total Deaths ('000)	Home-less ('000)	Disaster type	Total Damage (B US $)
1963-09-07	12.5	24.2	Flood	13.2
1979-05-21	7.8	22.5	Earthquake	2.5
1981-03-11	18.2	26.8	Storm	26.7
2001-11-09	9.3	23.7	Earthquake	3.3
2005-01-05	0.7	25.1	Volcanic activity	6.9
2008-06-27	23.0	28.5	Earthquake	32.9
2021-08-13	6.1	21.4	Flood	9.5

Datasets: Features and Target Variables

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{n}$ be a dataset

- x_i is a p-dimensional feature vector (independent variables)
- y_i is a is the respective target variable (dependent variable)

Example:

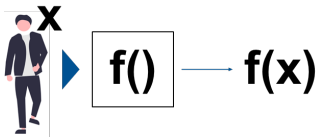
Each data points has an unordered collection of features ← if the structure is essential, it needs to be embedded

e.g., x = {height=180cm, hobbies={"tennis","coding"}, hometown="Munich"}

Feature

Realization of the feature "height"

y = "student" ← *Target variable*



Different feature types: categorical, ordinal, and numerical features

Feature Types I – Categorical Features

Categorical Features	Example
Each instance falls into one category of a set of **categories**	**colors**: {yellow, red, green, blue}
How are categories defined? 1. Categories are mutually exclusive 2. No natural ordering of the categories 3. Thus, the only meaningful operation is equality testing ($=, \neq$)	yellow $\neq$ red
What are challenges? In real-world datasets: synonyms It is crucial to analyze whether categories can be subsumed under an umbrella term	golden == yellow? azure == blue?
Further examples of categorical data?	marital status, brand, …
Example from our NASA earth data	EARTHDATA `df['Disaster Type'] = df['Disaster Type'].astype('category')`

Feature Types II – Ordinal Features

Ordinal Features	Example
Instance falls into a set of **categories with a natural ordering**	**Likert scale**: {strongly disagree, disagree, neutral, agree, strongly agree} **education level**: {none, school, university, post-grad}
How are the categories treated? 1. Categories are encoded as numbers to preserve their ordering 2. It is meaningful to compare values ($<, >, =$) 3. However, the "difference" between these encoded numbers should not be measured, added, or multiplied	
How do you differentiate ordinal from categorical data? Can be challenging sometimes. Does {none, school, university, post-grad} have a natural ordering?	**language ability**, **income level**
Example from our NASA earth data	 Aid Contribution: {None, Delayed, Immediate}

Feature Types III – Numerical Features

Numerical Features

Instances are **integers or real numbers** (difference relevant)

How are numerical features treated?

1. It is meaningful to add, multiply, compute mean or variance (etc.) of target variables

2. Often variables are **normalized** in order **to reach zero mean and unit variance**: x' = (x - mean) / std. dev.

   alternatively: **Normalization on [0,1]**: x' = (x - min) / (max - min)

3. Before normalization, handle extreme values (outliers)

   e.g.: **height**: {180, 172, 190, 120, 175, 2000} [cm]

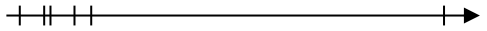
Example from our NASA earth data

Example

height: {180, 172, 190, 120, 175} [cm]

age, **stock price**, **property value**

Unit variance: **standard deviation** and **variance** of a sample will tend towards 1 as sample size tends towards infinity.



EARTHDATA

number of death (*Integer*), number of homeless (*Integer*), total damage (*Float*)

Input and target features can be formalized in matrixes.

Feature: an individual measurable property or characteristic of a phenomenon

We let $\boldsymbol{X}$ denote an $n \times p$ matrix whose (i, j) th element is $x_{i,j}$. That is,

$$\mathbf{X} = \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1p} \\ x_{21} & x_{22} & \dots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{np} \end{pmatrix}$$

Target: is the variable the model is trying to predict

We use y_i to denote i th observation of the variable on which we wish to make predictions. Hence, we write the set of all n observations in vector form as

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

Variables types and notations



Agenda.

- Terms Artificial Intelligence & Machine Learning
- Variables types and notations
- Descriptive univariate statistics
- Bivariate statistics
- Probability theory
- Bayes Theorem

We can investigate how often realizations of feature occur.

For simplicity, consider a dataset with a single feature $(p = 1)$ such that $\boldsymbol{X}$ denotes an $n \times 1$ matrix. Let k be the number of unique realizations of this feature and $m = 1, \ldots, k$:

Absolute frequency: $h(X = x_m)$

- with $0 \leq h(X = x_m) \leq n$ and $\sum_{m=1}^{k} h(X = x_m) = n$

Relative frequency: $f(X = x_m) = \frac{h(X = x_m)}{n}$

- with $0 \leq f(X = x_m) \leq 1$ and $\sum_{m=1}^{k} f(X = x_m) = 1$

`df['Disaster Type'].value_counts()`

EARTHDATA

Disaster Type	$h(X = x_m)$	$f(X = x_m)$
Flood	5551	0.34
Storm	4496	0.28
Earthquake	1544	0.1
Epidemic	1501	0.09
Landslide	776	0.05
…	…	…
Mass movement (dry)	48	0.0
Glacial lake outburst	2	0.0

`(df['Disaster Type'].value_counts()/len(df)).round(2)`

We can investigate how often realizations of feature occur.



And aggregate empirical cumulative distribution function.

Disaster Type	$h(X = x_m)$	$f(X = x_m)$
Flood	5551	0.34
Storm	4496	0.28
Earthquake	1544	0.1
Epidemic	1501	0.09
Landslide	776	0.05
…	…	…
Mass movement (dry)	48	0.0
Glacial lake outburst	2	0.0

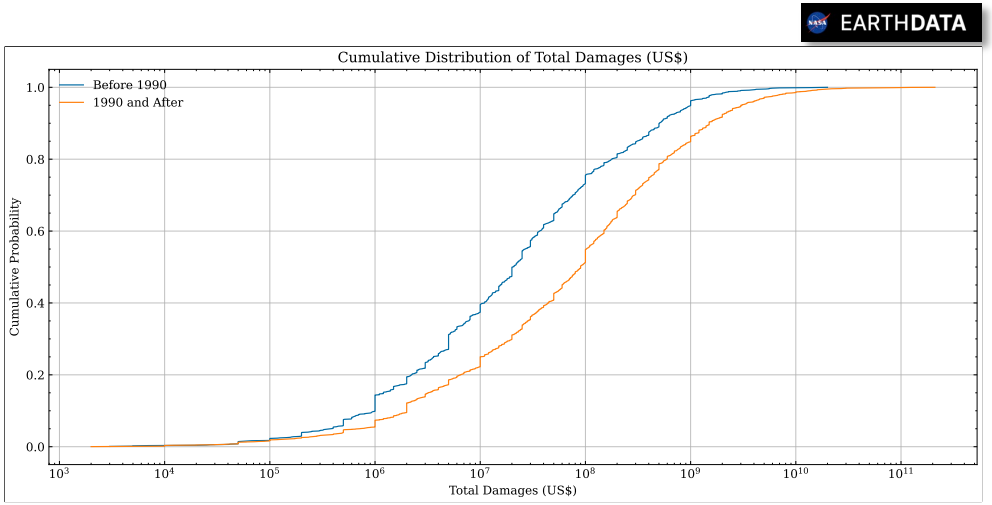


Figure 1.2: ecdf

Cumulative absolute frequency:

$$H(X = x_m) = \sum_{l=1}^{m} h(X = x_l)$$

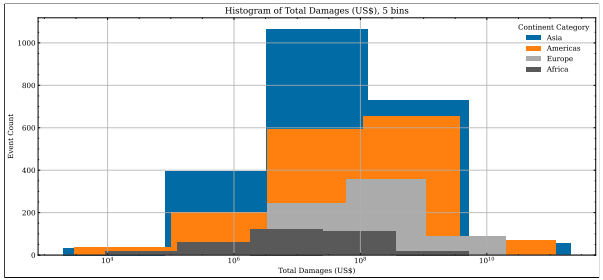
Cumulative relative frequency:

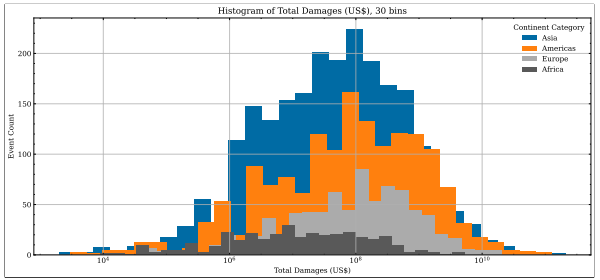
$$F(X = x_m) = \sum_{l=1}^{m} f(X = x_l)$$

Capital letter for cumulative distribution

? How do you interpret the graph shown?

Histograms are area proportional visualisation for distributions.





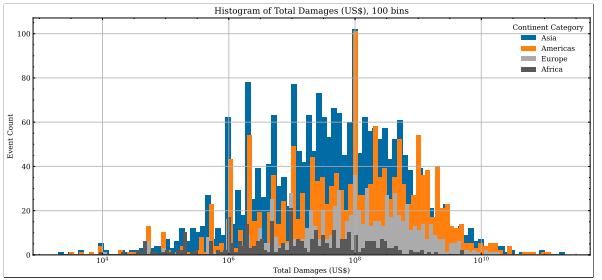
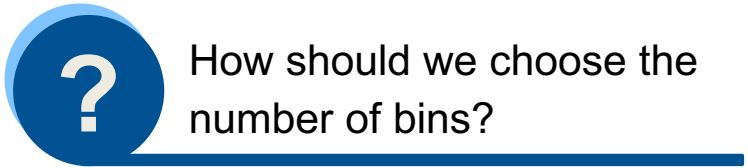


Figure 1.3: binning



We can report various characteristic to describe the distribution of our data.

Location parameter	Dispersion parameter
Arithmetic mean	Standard deviation
Median	Range
Mode	Interquartile range
Quantile	

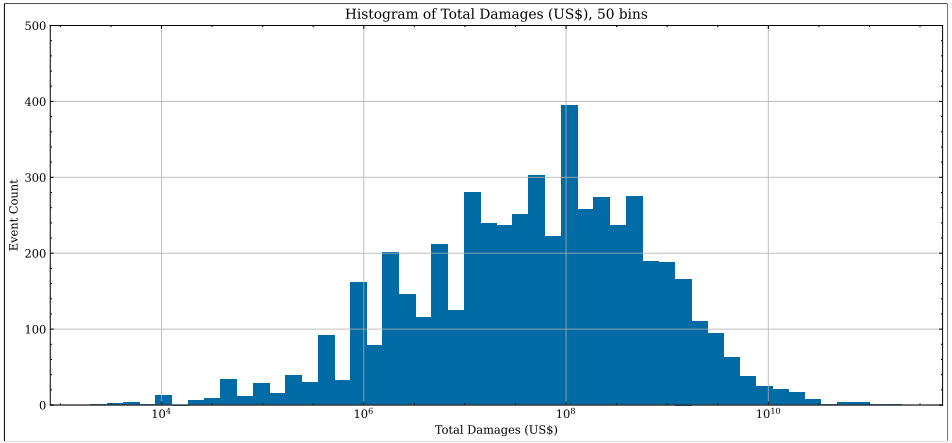


Figure 1.4: historgram measures

Which location and dispersion parameters do you know?

We can report various location parameters to describe the distribution of our data.

Location parameter	Formula
Mean	$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$
Median	Let $\mathcal{X} = \{x_i\}_{i=1}^{n}$ be an order set, such that: $x_1 \leq x_2 \leq \cdots \leq x_n$ $\tilde{x} = \begin{cases} x_{n/2} \text{if } x \text{ is odd} \\ (x_{n/2} + x_{n+1/2})/2 \text{ if } n \text{ is even} \end{cases}$
Mode	$\max_{m=1,\ldots,k} h(X = x_m)$
Quantile*	$F(X = x_m) = \sum_{l=1}^{m} f(X = x_l)$

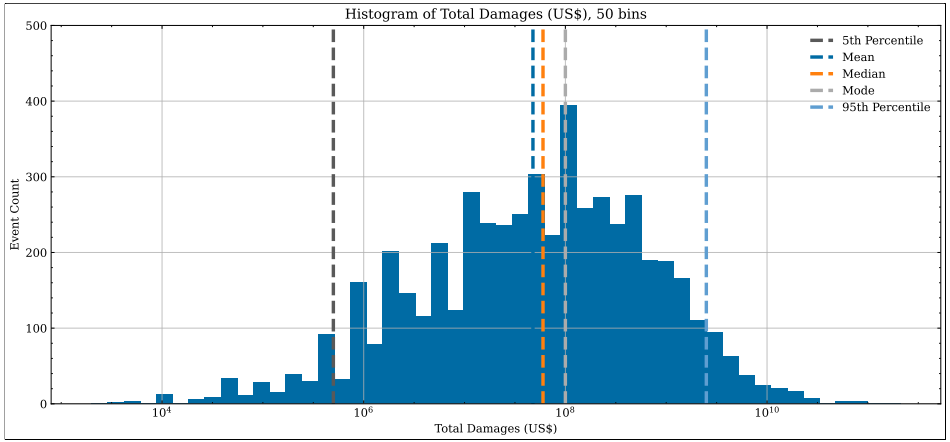


Figure 1.4: histogram measures

** Since F is a monotonically increasing function, it has an inverse, denoted by F^{-1}. For a given percentile α, we write $F^{-1}(\alpha)$ which identifies x_m, such that $F(X = x_m) = \alpha$.*

We can report various location parameters to describe the distribution of our data.



We can report various dispersion parameters to describe the distribution of our data.

Dispersion parameter	Formula
Standard deviation	$\sigma = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}$
Range	Let $\mathcal{X} = \{x_i\}_{i=1}^{n}$ be an order set, such that: $x_1 \leq x_2 \leq \cdots \leq x_n$ $x_n - x_1$
Interquartile range	$F(X = x_{0.75}) - F(X = x_{0.25})$

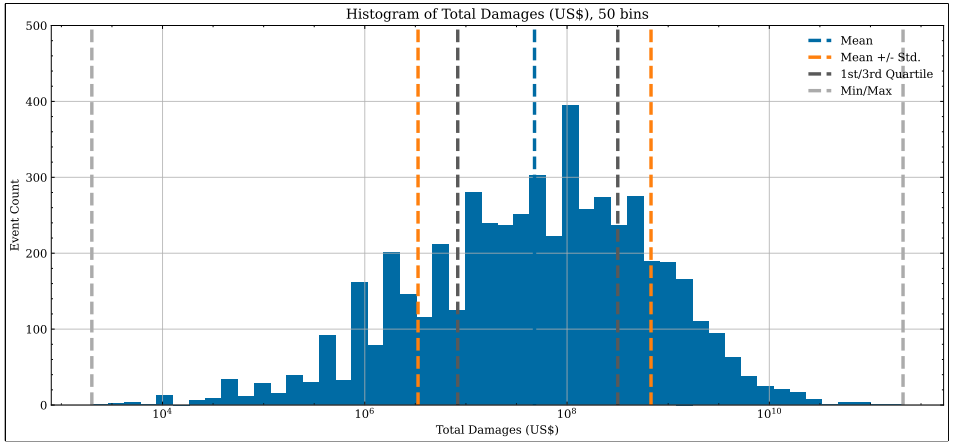


Figure 1.4: historgram measures

We can report various dispersion parameters to describe the distribution of our data.

matplotlib [1]

```
     Q1-1.5IQR   Q1   median  Q3   Q3+1.5IQR
                  |-----:-----|
  o      |--------|     :     |--------|    o  o
                  |-----:-----|
flier             <----------->            fliers
                        IQR
```

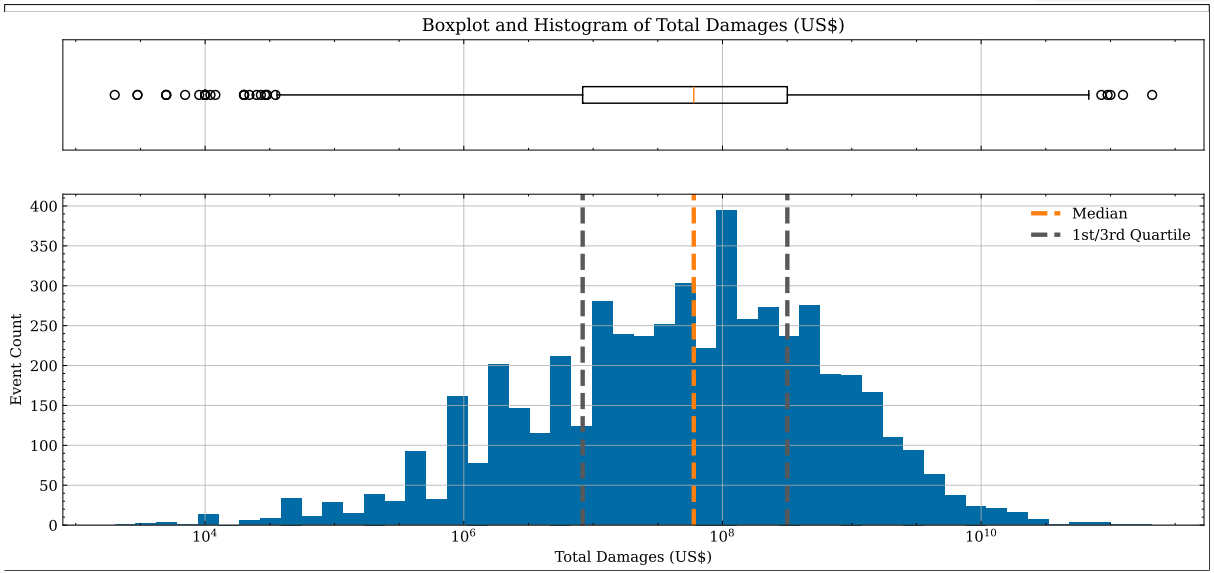


Figure 1.5: boxplot dispersion

[1] https://matplotlib.org/stable/api/_as_gen/matplotlib.pyplot.boxplot.html

We can report various characteristics to describe the distribution of our data.



Guidelines on when our location and dispersion parameters are useful.

	Categorical	Ordinal	Numerical Integer	Numerical Continuous
Location parameter				
Mean				
Median				
Mode				
Quantile				
Dispersion parameter				
Standard deviation				
Range				
Interquartile range				

Probably fine

challenging

no way



How about categorical variables that we transformed to numeric representation?

Credits to Prof. Dr. Melanie Schienle, Statistical Methods and Econometrics, KIT

Guidelines on when our location and dispersion parameters are useful.

Example	Variable	Useful parameters	Intuition
Which color of car is most common in the city?	Car color (categorical)	Mode	Colors have no numerical meaning, so only the most frequent category (mode) captures central tendency. Mean and median make no sense here.
What is the satisfaction level of students with the course offerings at TUM?	Satisfaction rating (1–5 Likert scale, ordinal)	Median or quantiles	The order carries meaning, but distances between levels are not equal. The median tells you the "middle" satisfaction level, while the mean would incorrectly assume equal intervals between ratings.
What is the number of ChatGPT visits per user per week?	Number of visits (discrete numerical)	Mean, median, range, standard deviation	Although discrete, these counts are numerical. Mean and standard deviation summarize central tendency and variability effectively when the distribution is not too skewed.
What is the number of children per household?	Number of children (discrete numerical, skewed)	Median or mode, interquartile range	The mean is distorted by few large families, so it really depends on the question at hand. The median better represents the "typical" case, and the interquartile range captures spread without being overly influenced by outliers.
What is the temperature in Munich in July like?	Temperature in °C (continuous numerical)	Mean, standard deviation	Temperature is continuous and approximately symmetric, so the mean and standard deviation provide a concise and informative summary of center and spread.
What is the income of individuals in MUC?	Annual income (continuous, right-skewed)	Median, interquartile range, quantiles	Income distributions are heavily right-skewed; the mean would overrepresent extreme earners. The median gives a robust central measure, and the IQR highlights variability among the majority. If the income includes stock options, the variable becomes continues beyond the decimal point of cents.
Which age groups are overrepresented in Germany?	Age group (ordinal categories)	Median (if ordered), quantile boundaries	Though categorical, the groups have order; therefore, ordinal logic applies. A mean would be meaningless, but median or quantiles might be interpretable.

Agenda.

- Terms Artificial Intelligence & Machine Learning
- Variables types and notations
- Descriptive univariate statistics
- Bivariate statistics
- Probability theory
- Bayes Theorem

Bivariate statistics investigate distributions among two variables.

Consider a dataset with two features $(p = 2)$ such that $\boldsymbol{X}$ denotes an $n \times 2$ matrix. Let k be the number of unique realizations of our first feature and let $x_{i,j}^1$ refer to the first feature of $x_{i,j}$ and X^1 to the matrix notation. Also let r be the number of unique realizations of our second feature $x_{i,j}^2$:

Joint absolute distribution:

$h_{m,s} = h(x_m^1, x_s^2)$

Joint relative distribution:

$f_{m,s} = f(x_m^1, x_s^2)$ with $m = 1, \dots, k$ and $s = 1, \dots, r$

Marginal absolute distribution:

$h_{m\cdot} = \sum_{s=1}^{r} h_{m,s}$

$h_{\cdot s} = \sum_{m=1}^{k} h_{m,s}$

Marginal relative distribution:

$f_{m\cdot} = \sum_{s=1}^{r} f_{m,s}$ with $m = 1, \dots, k$

$f_{\cdot s} = \sum_{m=1}^{k} f_{m,s}$ with $s = 1, \dots, r$

EARTHDATA

$h_{m,s}\ (f_{m,s})$	Earth-quake	Flood	Storm	Other	Total $h_{m\cdot}\ (f_{m\cdot})$
Africa	74 (0.0)	1147 (0.07)	290 (0.02)	1435 (0.09)	**2946 (0.18)**
Americas	314 (0.02)	1275 (0.08)	1442 (0.09)	940 (0.06)	**3971 (0.25)**
Asia	903 (0.06)	2303 (0.14)	1884 (0.12)	1400 (0.09)	**6490 (0.4)**
Europe	197 (0.01)	668 (0.04)	542 (0.03)	590 (0.04)	**1997 (0.12)**
Other	56 (0.0)	158 (0.01)	338 (0.02)	170 (0.01)	**722 (0.04)**
Total $h_{\cdot s}\ (f_{\cdot s})$	**1544 (0.1)**	**5551 (0.34)**	**4496 (0.28)**	**4535 (0.28)**	**16126 (1.0)**

Bivariate statistics investigate distributions among two variables.

Derive the **Conditional distribution** by putting the **Joint distribution** and the **Marginal distribution** into context:

Conditional distribution:

$$f(x_m^1 | x^2 = x_s^2) = \frac{f(x_m^1, x_s^2)}{f_{\cdot s}} = \frac{h(x_m^1, x_s^2)}{h_{\cdot s}}$$

So, we can answer questions like: *We observe disaster reporting in Asia, what's the probability of the disaster being a Flood?*

$$f(x^2 = Flood | x^1 = Asia) = \frac{f(x^2 = Flood, x^1 = Asia)}{f(x^1 = Asia)} = \frac{0.14}{0.4} = 35\%$$

EARTHDATA

$h_{m,s}$ ($f_{m,s}$)	Earth-quake	Flood	Storm	Other	Total $h_{m\cdot}$ ($f_{m\cdot}$)
Africa	74 (0.0)	1147 (0.07)	290 (0.02)	1435 (0.09)	**2946 (0.18)**
Americas	314 (0.02)	1275 (0.08)	1442 (0.09)	940 (0.06)	**3971 (0.25)**
Asia	903 (0.06)	2303 (0.14)	1884 (0.12)	1400 (0.09)	**6490 (0.4)**
Europe	197 (0.01)	668 (0.04)	542 (0.03)	590 (0.04)	**1997 (0.12)**
Other	56 (0.0)	158 (0.01)	338 (0.02)	170 (0.01)	**722 (0.04)**
Total $h_{\cdot s}$ ($f_{\cdot s}$)	**1544 (0.1)**	**5551 (0.34)**	**4496 (0.28)**	**4535 (0.28)**	**16126 (1.0)**

Bivariate statistics investigate distributions among two variables.

Variables are considered **independent** if:

$f(x_m^1|x^2 = x_s^2) = f(x_m^1|x^2 = x_l^2) = f(x_m^1)$ and $f(x_s^2|x^1 = x_m^1) = f(x_s^2|x^1 = x_q^1) = f(x_s^2)$

for all $s, l = 1, \dots, r$ and for all $m, q = 1, \dots, k$

Knowing about the realisation of one feature does not provide us any additional insights on the realisation of the other feature. In other words, the **Conditional distribution** is equivalent to the **Marginal distribution**.

And in this case: we can calculate the **joint distribution** by:

$f(x_m^1, x_s^2) = f(x_m^1) \times f(x_s^2)$

How do we calculate the joint distribution for **dependent** variables?

If variables are <u>not independent,</u> we get:

$f(x_m^1, x_s^2) = f(x_m^1) \times f(x_s^2|x^1 = x_m^1)$

Which variables would we like to be **independent**?
Which variables should be **dependent**?

EARTHDATA

$h_{m,s}\ (f_{m,s})$	Earth-quake	Flood	Storm	Other	Total $h_{m,\cdot}(f_{m,\cdot})$
Africa	74 (0.0)	1147 (0.07)	290 (0.02)	1435 (0.09)	**2946 (0.18)**
Americas	314 (0.02)	1275 (0.08)	1442 (0.09)	940 (0.06)	**3971 (0.25)**
Asia	903 (0.06)	2303 (0.14)	1884 (0.12)	1400 (0.09)	**6490 (0.4)**
Europe	197 (0.01)	668 (0.04)	542 (0.03)	590 (0.04)	**1997 (0.12)**
Other	56 (0.0)	158 (0.01)	338 (0.02)	170 (0.01)	**722 (0.04)**
Total $h_{\cdot,s}\ (f_{\cdot,s})$	**1544 (0.1)**	**5551 (0.34)**	**4496 (0.28)**	**4535 (0.28)**	**16126 (1.0)**

Bivariate statistics investigate distributions among two variables.



Agenda.

- Terms Artificial Intelligence & Machine Learning
- Variables types and notations
- Descriptive univariate statistics
- Bivariate statistics
- Probability theory
- Bayes Theorem

From descriptive statistics to probabilities.

Moving from what we observed in the past and reported via descriptive statistics, we eventually want to talk about the future and derive what we expect to happen. To do so statistician agreed on the term **Probability**, formalized as:

$$P(A) = \lim_{n \to \infty} f_n(A) \in [0,1]$$

With A referring to a so-called **event**, e.g. an *Earthquake in Asia*. There are different interpretations of an probability:

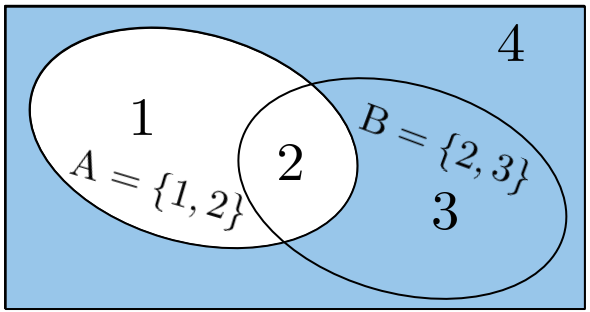
Frequentist: *The probability of an outcome is the proportion of times the outcome would occur if we observed the random process an infinite number of times.*

Bayesian: *Probability is a subjective degree of belief: For the same event, two separate people could have different viewpoints and so assign different probabilities.*

EARTHDATA

$h_{m,s}\,(f_{m,s})$	Earth-quake	Flood	Storm	Other	Total $h_{m\cdot}(f_{m\cdot})$
Africa	74 (0.0)	1147 (0.07)	290 (0.02)	1435 (0.09)	**2946 (0.18)**
Americas	314 (0.02)	1275 (0.08)	1442 (0.09)	940 (0.06)	**3971 (0.25)**
Asia	903 (0.06)	2303 (0.14)	1884 (0.12)	1400 (0.09)	**6490 (0.4)**
Europe	197 (0.01)	668 (0.04)	542 (0.03)	590 (0.04)	**1997 (0.12)**
Other	56 (0.0)	158 (0.01)	338 (0.02)	170 (0.01)	**722 (0.04)**
Total $h_{\cdot s}\,(f_{\cdot s})$	**1544 (0.1)**	**5551 (0.34)**	**4496 (0.28)**	**4535 (0.28)**	**16126 (1.0)**

Recap: General Set Notation

Set Notation	Meaning	Venn Diagram	
$\mathrm{A} \cup \mathrm{B}$	Everything in both sets *Union*		$\{1, 2, 3\}$
$\mathrm{A} \cap \mathrm{B}$	What is common in both sets *Intersection*		$\{2\}$
$\overline{\mathrm{A}}$, A' or $\neg \mathrm{A}$	Everything not in that set *Complement*		$\{3, 4\}$

Recap: Probability Notation

Probability is the likelihood of an event occurring; Probabilities range from $[0, 1]$

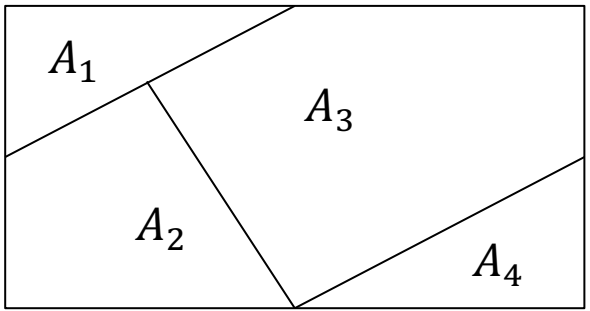
We denote the probability of an event as $\mathrm{P}(\text{desired event}) = \dfrac{\text{Number of desired events}}{\text{Total number of events}}$

$\mathrm{P}(\mathrm{A})$	Event A	The probability of event A happening
$\mathrm{P}(\overline{\mathrm{A}})$	Complement	The probability of event A not happening
$\mathrm{P}(\mathrm{A} \cup \mathrm{B})$	Union	The probability of event A or event B happening
$\mathrm{P}(\mathrm{A} \cap \mathrm{B})$	Intersection	The probability of event A and event B happening

Conditional Probability

Probability of event A happening given that event B has occurred: $\mathrm{P}(\mathrm{A}|\mathrm{B})$

Recap: Pairwise disjoint

Set Notation	Meaning	Venn Diagram
$A_i \cap A_j = \emptyset$ for all $i \neq j$	Nothing is in two sets. *Pairwise Disjoint*	$S = \{A_1, A_2, \dots, A_n\}$ 
$\bigcup_{i=1}^{n} A_i = S$	Everything is in one set. *Complete*	
$P(S) = 1$		

Conditional Probability

Independent Events

If two events do not influence each other, the probability that A occurs stays the same, independent of B

$$P(A|B) = P(A)$$

Probability of A given B

$$P(A \cap B) = P(A)\,\mathrm{P(B)}$$

Probability of A and B

Dependent Events

Product rule for **conditional probability**, to express dependent events, i.e., events influence each other

$$P(A|B) = P(A \cap B)/P(\mathrm{B})$$

Probability of A given B

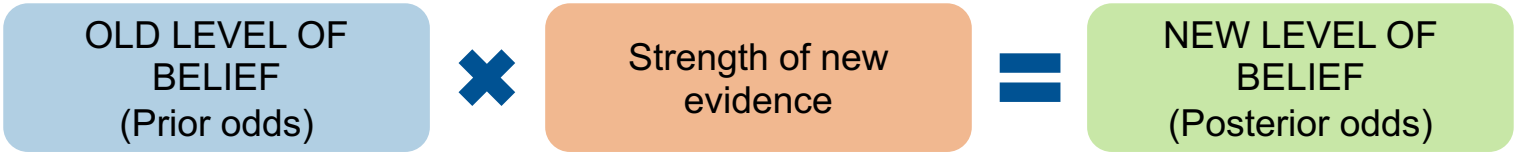
$$P(A \cap B) = P(A|B)\,\mathrm{P(B)} = P(\mathrm{B|A})\,\mathrm{P(A)}$$

Probability of A and B determined by probability of A given B times probability of B

Agenda.

- Terms Artificial Intelligence & Machine Learning
- Variables types and notations
- Descriptive univariate statistics
- Bivariate statistics
- Probability theory
- Bayes Theorem

Bayes Theorem - Motivation



The motivation behind Bayes' theorem is to **update or revise probabilities based on new evidence or observations**.

It provides a way to calculate the probability of a hypothesis or event, given the probability of the evidence, by incorporating prior knowledge or beliefs about the hypothesis and updating them with new data, leading to **more accurate and informed predictions**.

Bayes Theorem

Posterior Probability, conditional probability of A given B

Conditional probability of B given A

Probability of A

$$\mathrm{P(A|B)} = \frac{\mathrm{P(B|A)\,P(A)}}{\mathrm{P(B)}}$$

Probability of B

Bayes Theorem provides a way to update the probability of a hypothesis based on new evidence. The theorem considers both the initial probability of the hypothesis and how likely the new evidence is.

Bayes Theorem – Visual Example

Your friend Alex is very tidy and loves organizing → what is the probability that Alex' job is farmer vs the probability that Alex works as a librarian?

$$\mathrm{P}(\text{librarian} \mid \text{tidy}) = \mathrm{P}(\text{📚} \mid \text{🧹}) = 80\%$$

$$\mathrm{P}(\text{📚} \mid \overline{\text{🧹}}) = 20\%$$

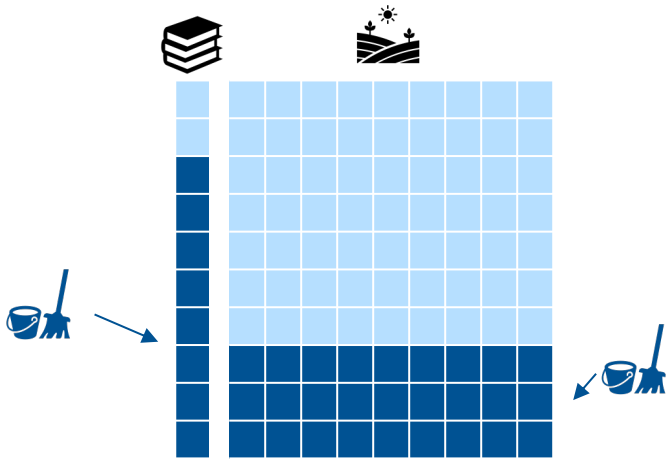
$$\mathrm{P}(\text{🌾} \mid \text{🧹}) = 30\%$$

$$\mathrm{P}(\text{🌾} \mid \overline{\text{🧹}}) = 70\%$$

Observing the jobs of 100 randomly selected people:

$$\#\,\text{📚} = 10$$

$$\#\,\text{🌾} = 90$$



$$\mathrm{P}(\text{📚} \mid \text{🧹}) = \frac{\#\,\text{🧹}\text{📚}}{\#\,\text{🧹}\text{📚} + \#\,\text{🧹}\text{🌾}} = \frac{10 \cdot 0.8}{10 \cdot 0.8 + 90 \cdot 0.3} \approx 0.3$$

$$\mathrm{P}(\text{🌾} \mid \text{🧹}) = \frac{\#\,\text{🧹}\text{🌾}}{\#\,\text{🧹}\text{📚} + \#\,\text{🧹}\text{🌾}} = \frac{90 \cdot 0.3}{10 \cdot 0.8 + 90 \cdot 0.3} \approx 0.7$$